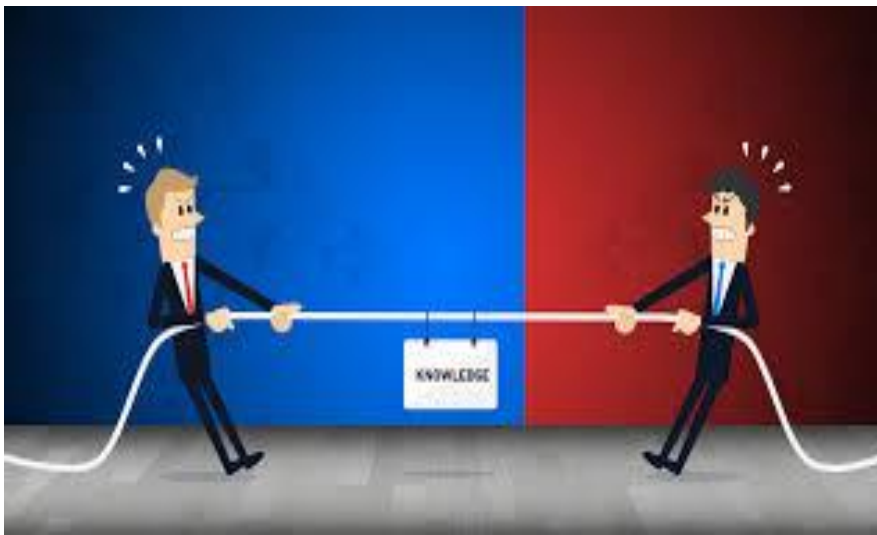


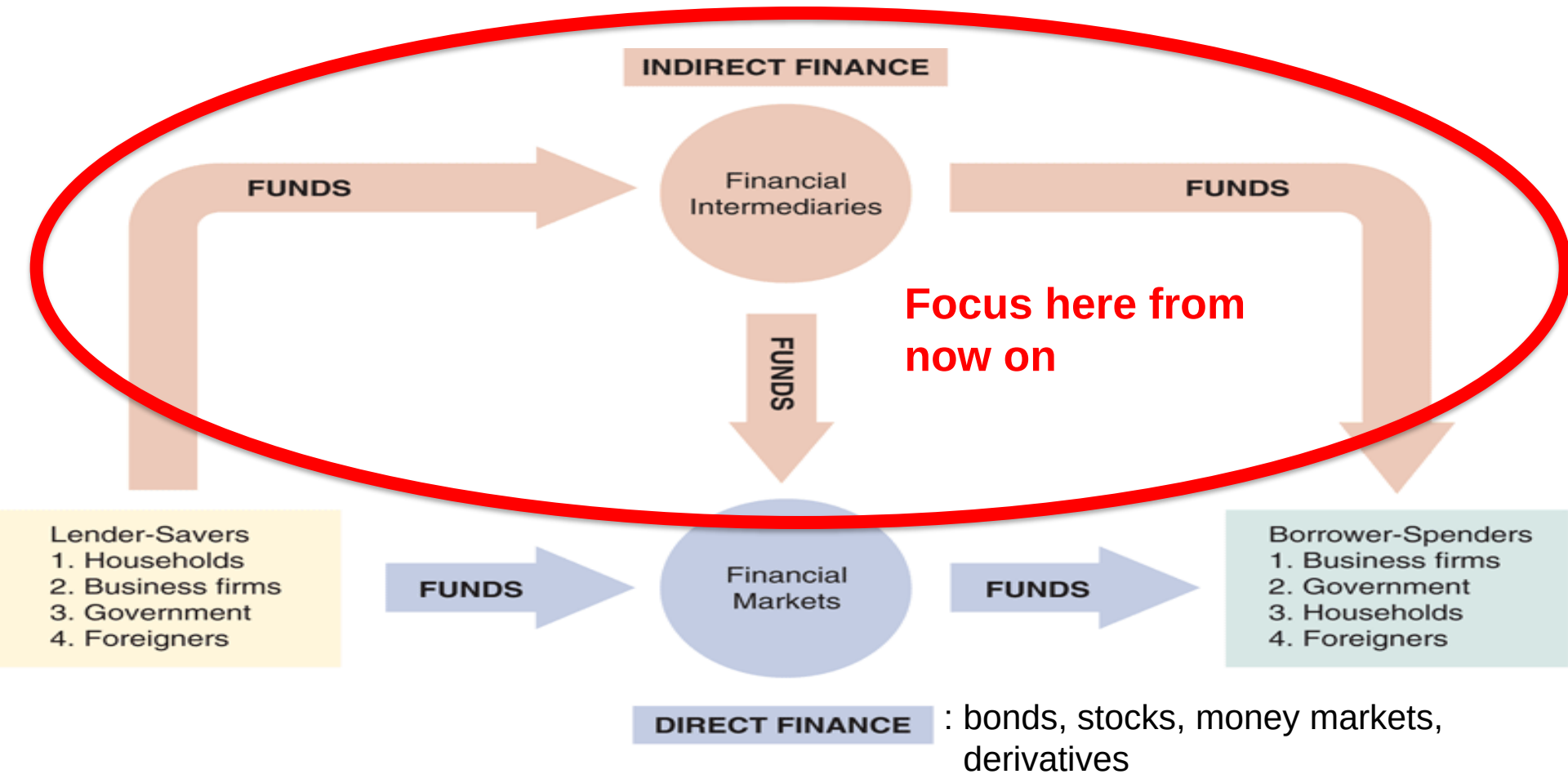
Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

Prof. Francesco Bripi

Why Do Financial Institutions Exist? (Chapter 7)



Part II: Indirect Finance

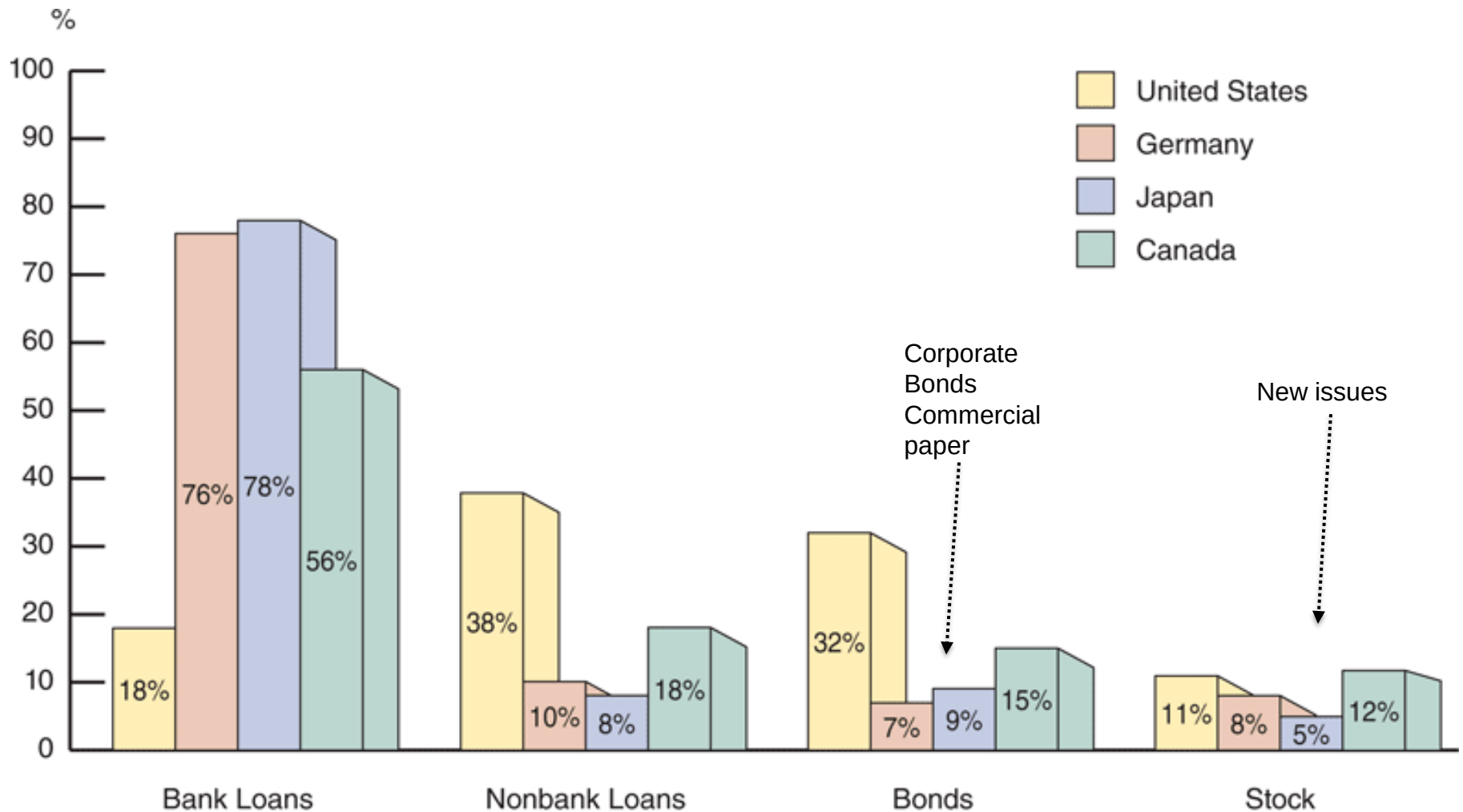


The arrows show that funds flow from lender-savers to borrower-spenders via two routes: *direct finance*, in which borrowers borrow funds directly from financial markets by selling securities, and *indirect finance*, in which a financial intermediary borrows funds from lender-savers and then uses these funds to make loans to borrower-spenders.

Part II: Indirect Finance

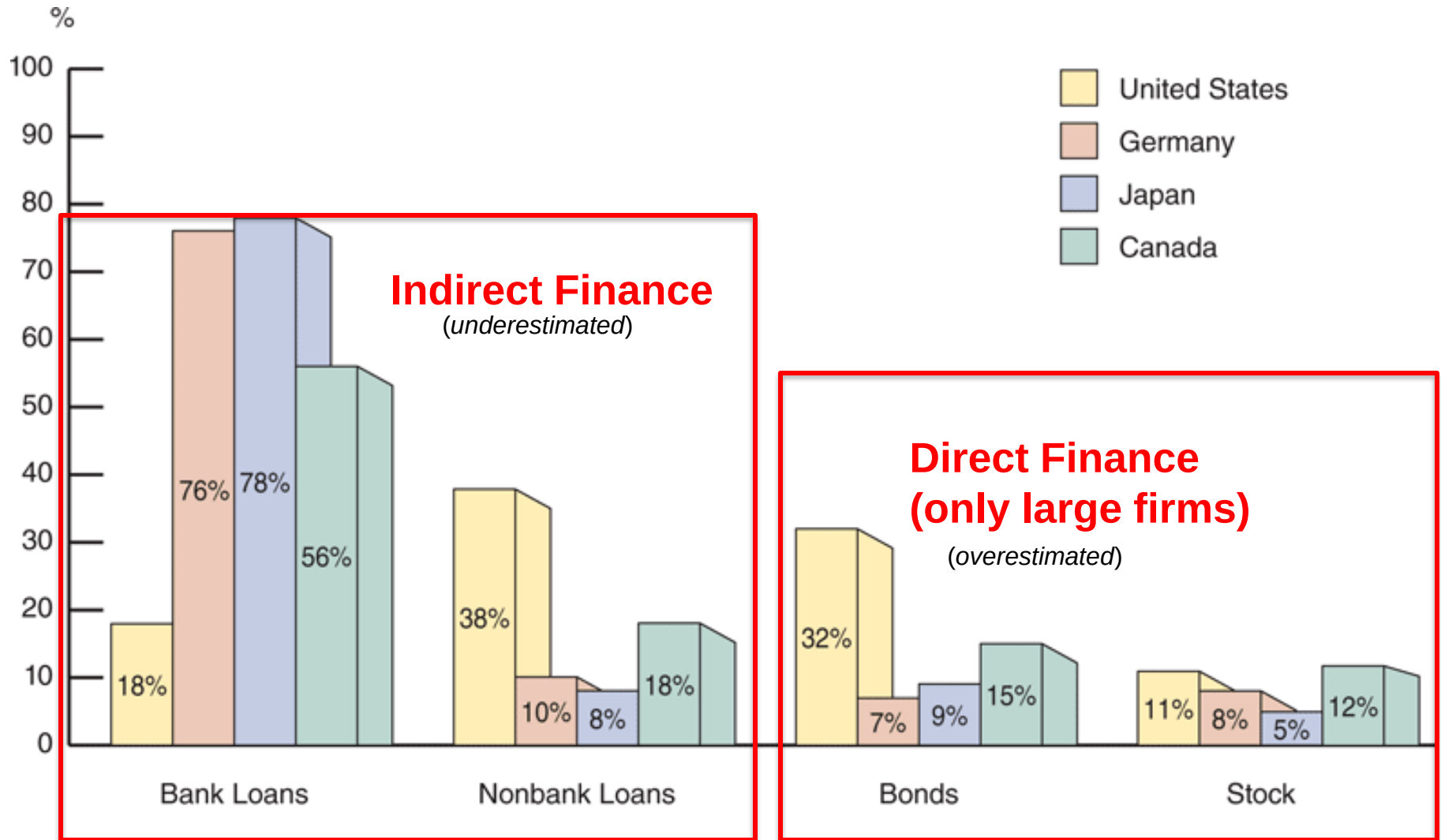
- ✓ Today we will look at **Financial Intermediaries** in general: functions, advantages...
- ✓ Next two lectures: **Commercial Banks** and **Central Banks**
- ✓ Then move on to: **Mutual Funds** and **Investment Banks**
- ✓ After, some topics regarding financial institutions in general: (**Risk Management, Regulation** and **Financial Crises**)
- ✓ Last two lectures with me
 - Financial Crises
 - Review and exercises
- ✓ Special lecture (3rd December)
 - Dr. Giorgio Gobbi, Senior Director at Bank of Italy,
 - long experience on Financial Stability and Fintech

Sources of External Finance for Non-Financial Business



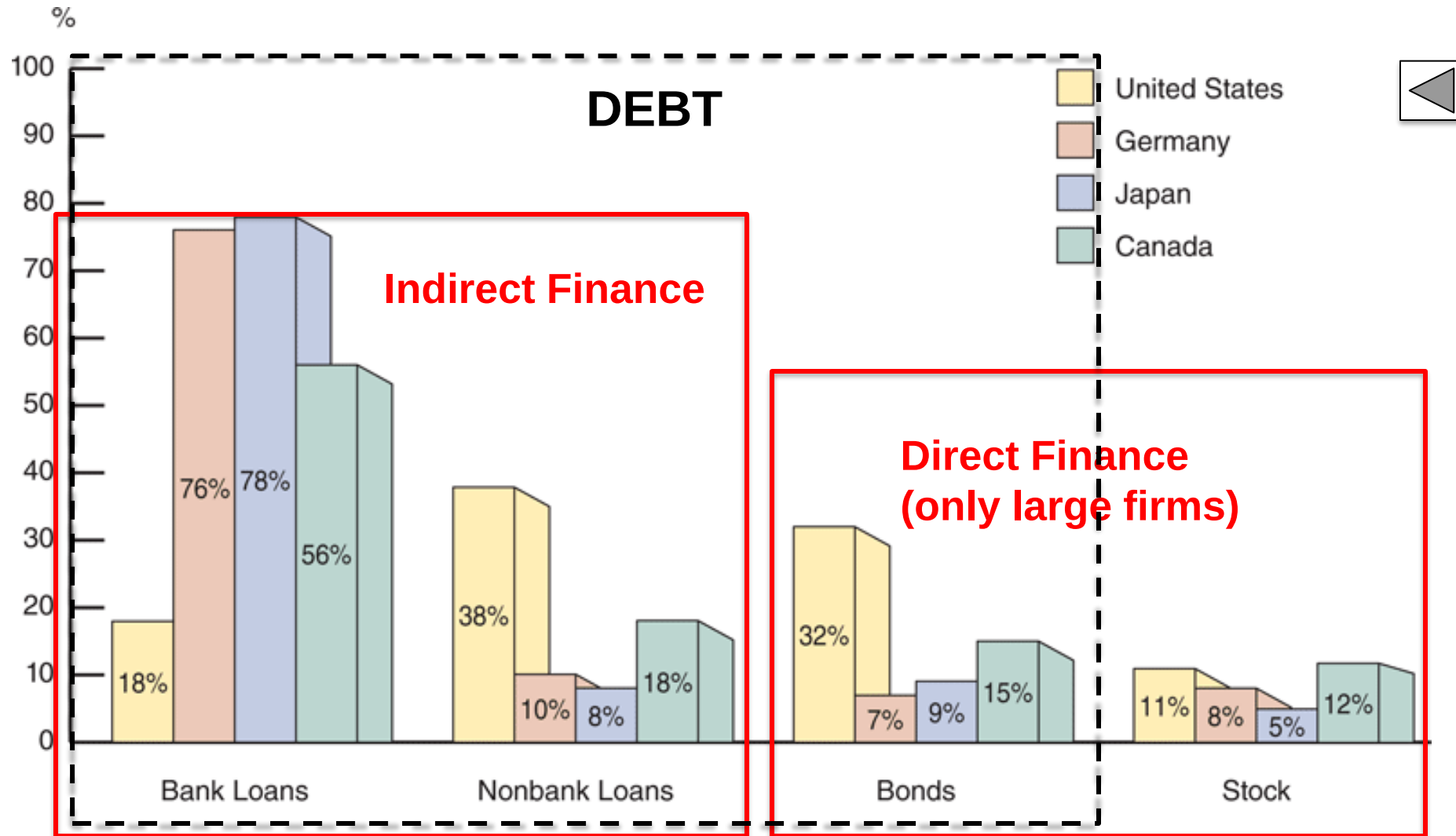
Source: Andreas Hackethal and Reinhard H. Schmidt, "Financing Patterns: Measurement Concepts and Empirical Results," Johann Wolfgang Goethe-Universitat Working Paper No. 125, January 2004. The data are from 1970–2000 and are gross flows as percentages of the total, not including trade and other credit data, which are not available.

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Outline of today

We go back to the foundations of financial institutions

1. Why do financial institutions exist?
2. Transaction costs
3. Two sides of Asymmetric Information (Asy Info)
 - Adverse Selection (Asy Info ex-*ante*)
 - Moral Hazard (Asy Info ex-*post*)
4. Some well-known conflicts of interest in financial industry

Motivation

Why do Financial Institutions (FI) exist?

- ✓ Two primary functions of FI (indirect finance):
 1. Lower **Transaction Costs**
 2. Reduce **Asymmetric Info**
- ✓ *Reason for their existence*
 - at least in principle, FI are *better equipped* than financial markets (direct finance) to solve these problems

By doing 1+2, FI engage in:

- ✓ **Risk sharing**
 - better ability to use portfolio *diversification* techniques
 - risk sharing due to lower *transaction* costs
- ✓ **Asset Transformation**: transform assets into more attractive ones:
 - *maturity* transformation (long maturities)
 - *denomination* transformation (large minimum investments)

Transaction Costs

- ✓ **Transactions costs** influence the financial structure
- ✓ Certain types of investments only available in **large denominations**
- ✓ For a *small (number of) transactions*, commissions and fees represent a high percentage of investments
- ✓ As a result, small individual investments are not well diversified (bear greater risk)

Transaction Costs

FI reduce transactions costs by:

- a) **Economies of Scale**: average cost (per \$ of investment) decreases as the size (*scale*) of transaction increases
 - consider fixed cost in buying shares
 - FI's can deliver lower cost per transaction simply by making so many financial transactions

- b) **Economies of Scope**: average cost decreases as number of products offered increases
 - transaction cost for a single transaction decreases if FI offers more products. Examples:
 - credit card & bank account & mortgage at same institution
 - online banking & trading platform (the cost of providing additional services via an existing online banking platform is negligible)

Economies of scale and Risk Sharing

Very broad *diversification* can be attained via a FI at *lower transaction costs*!

- ✓ Suppose you want to diversify \$2,000 over 60 different stocks
- ✓ Stock-broker charges \$4.59 flat for trade on each stock
- ✓ Your transaction cost would be very high if pay for each different trade
 - you'd pay \$275.4, i.e. **14%** of your investment in fees!
 - diversification too costly: not feasible!
- ✓ If instead 100 investors were to give \$2,000 each to an intermediary. The intermediary would diversify those \$200,000 over 60 different stocks (via the same stock-broker)
 - they'd pay all together still \$275.4 (only 60 trades overall!)
 - **0.14%** of their individual contribution in transaction fees!

Economies of scale and Asset transformation

Denomination Transformation

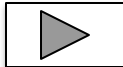
- ✓ Many securities are in principle very interesting but come in large denominations
 - one single class A share of Warren Buffet's Berkshire Hathaway Inc. costs **\$736,653** (as of mid Oct. 2025)
 - most *commercial paper* usually comes in denominations of **\$100k** to **\$250k**
- ✓ You could buy 1, 2, 3, . . . (but no less than 1)
- ✓ By investing via a FI ...
 - ...you can nevertheless invest small amounts in securities that have *large* denominations
 - ex of FI: bank or mutual fund

Facts: Direct vs Indirect Finance

Transaction costs **cannot** explain some facts!

Throughout the world, non-financial businesses mostly use **indirect finance** as a source of external funds (*i.e.* all those funds that do not come from the business itself through retained earnings)

FACT 1 **Small firms** *typically* do **not** issue *bonds+stocks*

FACT 2 There is more debt (bonds+loans) than equity around the world 

FACT 3 Bank loans often require collateral (*secured*)

WHY?

Main reason: **Asymmetric Information**

Asymmetric Information

- ✓ *Definition*: there is asymmetric info when one counterparty lacks crucial information about another party, impacting decision-making
- ✓ The market may *break-down* in the presence of asymmetric info (market failure)
- ✓ Agency theory: analysis of how Asy Info affects economic behavior
- ✓ Two types of Asymmetric Info:
 - **Adverse selection**
 - and
 - **Moral hazard**

Asymmetric Information

Adverse Selection

1. one party that knows more about the transaction most likely produce the undesirable (*adverse*) outcome
 2. Asy info exists before transaction occurs (**ex-*ante***)
-
- ✓ Famous “*lemon*” market problem (Akerlof) in the market for secondhand cars
 - good or bad (lemon) car
 - good or bad borrower

Adverse Selection

Adverse Selection

- ✓ *Before* agreeing to a transaction (ex-*ante*), one party has more information. 4 ex.:
 1. A health *insurance* does not know exact medical condition of a potential customer
 2. A buyer of a *used car* does not know its quality
 3. A bank does not know the creditworthiness of a potential *borrower*
- ✓ How will each of these respond?
 1. Health insurance will charge premiums
 2. Buyer will ask a to buy a used car
 3. Bank will charge a interest rate

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Response: best types may **withdraw** from the market
... *Adverse Selection*

1. Healthy clients do **not** want to pay too high premiums
2. Good used cars would be **undervalued**
3. Best borrowers do **not** want to pay too high interest rate

Exercise: Adverse Selection

You want to buy a corporate bond.

There are three bonds offered:

- 1 high-quality value \$ 120
- 1 medium-quality value \$ 90
- 1 poor quality value \$ 60

You can't tell which is which but assume also the bond dealer does **not know** the quality of his bonds (*No Info*)

Q: if she offers a bond for sale, how much are you willing to pay for it?

Exercise: Adverse Selection

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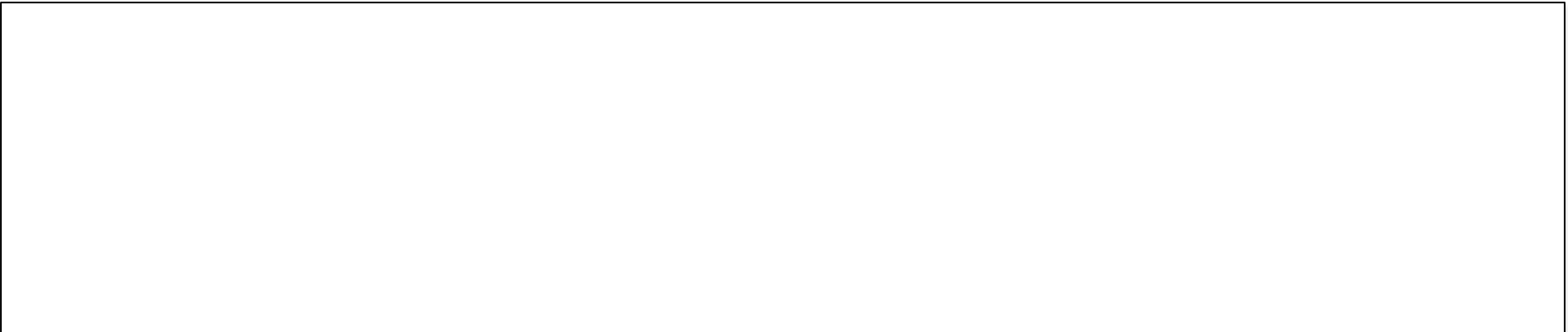
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A: willing to pay **\$60** at most. He'll sell you the junk no matter what:

- if you paid \$90 or \$120, he'd sell you the junk and earn \$30 or \$60
- since you know it, you willing to pay only \$60

Adverse Selection

✓ Reputation

- Previous example assumes that there are no reputational issues: if there are repeated interactions, the bond holder will not sell me a *lemon* the first time

✓ Firm size matters

- **Large well-established firms** should have less asymmetric info with financial markets (direct finance): reputation, collateral, etc...
- SME's very opaque w.r.t. information

✓ Hence large firms can **more easily issue stocks and bonds**

- less likely that the investors would buy a *lemon* (*pecking order hypothesis*)

✓ Explains FACT 1: small firms have hard time issuing stocks and bonds directly (Adverse Selection)

Solutions to Adverse Selection

How do you **solve** Adverse Selection problems?

1. Private Collection of information

- health examination before selling insurance
- Credit Rating Agencies (CRA): S&P, Moody's, Fitch, etc.
- credit scores to get credit card
- ✓ but private production of info creates a *free riding* problem:
 - ✓ once info is out in the market, hard to keep it, other can use this info for free, by simply observing your actions

2. Government Regulation

- reduce Asy Info by having true company's information revealed
- Auditors check financial statements (does not always work: see *Arthur Andersen* in the Enron scandal or Parmalat)

Solutions to Adverse Selection

3. Screening ex-ante from FI (especially banks)

- *private* collection of information
- but rather than selling info (like rating agencies)
- FI make private loans (*i.e.* non-traded) that other investors cannot observe
- banks also repeatedly lend to same businesses (*soft information*)

3. Collateral / Net Worth

- idea: ask ex-ante to borrowers to have some “*skin in the game*”
- collateral examples:
 - house for a mortgage
 - car for car loan
 - firms use Net Worth ($NW = \text{Assets} - \text{Liabilities}$) to finance investment

Is Adverse Selection Enough?

- ✓ Adverse selection can explain why
 - FACT 1: *small firms* are shut out of capital markets and use banks and
 - FACT 3: why *bank loans* are usually collateralized

- ✓ One point still remains unexplained (FACT 2):
 - why so **little equity** as a source of external finance?
or say
 - why **so much debt** (bonds+loans)?

Moral Hazard can explain that

Moral Hazard

Moral Hazard is a situation where one party has the incentive (*hazard*) to engage in undesirable (*immoral*) *activities* after the transaction has occurred (**ex-post**)

Examples:

- once people get insurance, they become careless
- once the loan is obtained, entrepreneur uses funds to buy perks (Ferrari) than the one approved (Machinery)
- your investor uses your savings for a riskier project than the one promised
- employee can become lazy right after a promotion
- customers leaving full plates at all-you-can-eat sushi

Problem is the unknown “**action**” of counter party

Case of MH: Principal-Agent Problem

Principal-Agent Problem

1. Result of separation
 - of **ownership** by “*many*” stockholders (*principals*)
 - from **control** by managers (*agents*)
 2. Problem: agents act in own rather than principals' interest
- ✓ Problem arises because the agents (e.g.: *managers*) *know more* about the company than principals (e.g.: *owners*)
 - ex.: managers make very risky investments
 - ex.: managers buy perks or steal company cash
 - ✓ Principals could **monitor** agents' activities but that's **costly**

Moral Hazard arises especially with equity contracts

Solutions to the Principal-Agent Problem (i.e. Moral Hazard with Equity)

1. Monitoring: ex-post screening

- ex.: shareholders can observe the level of effort of the manager
- problem with monitoring: it is costly (both time and money)
- FI are good at this
 - for example, **venture capitalists** do constant, active monitoring on the use of funds and receive equity share in start-ups that is *not tradable*

2. Government Intervention: enforce the laws on fraudulent behavior (hiding & stealing profits):

- imposing accounting standards
- efficiency of *civil justice* (optional readings)

Solutions to the Principal-Agent Problem (i.e. Moral Hazard with Equity)

3. Debt Contracts:

With equity Principal-Agent problem arises because equity is a claim on profits in all situations of the firm

- the principal (owners) needs to ensure that **no** profits are *diverted* and or *hidden away* by the agent ...
- ... but it requires monitoring (*costly state verification*)

✓ Debt contract can reduce monitoring costs

- only need to monitor in some states of the world:
 - *repayments past due*
 - default
- if the firm has high profits, the lender receives the fixed payment, as agreed, and does not need to monitor

✓ Important result in theoretical finance:

debt is the optimal security if moral hazard is present

Moral Hazard with Debt

Explains FACT 2: moral hazard can explain why debt (bonds and/or loans) is much more prevalent than equity

- ✓ But debt itself is not immune from moral hazard
 - Ex.: the entrepreneur (borrower), once the funds are obtained, has the incentive to take on riskier projects
 - Why? Because he only needs to pay a fixed interest: if he makes more, he can keep the rest
- ✓ To overcome moral hazard, debt contracts often have:
 1. **collateral** (incentive compatible debt contract)
 2. **covenants** (to prevent undesirable behavior)



FI usually specialized in monitoring
and require and enforce both 1 and 2

Conflicts of Interests

- ✓ FI's solve TC and AI problems, but sometimes their unique knowledge can also create other Asy Info problems (e.g.: m. h.):

“conflicts of interest”

- ✓ Mostly due to economies of scope in finance
 - Investment Banks (IBs) too optimistic: underwriting $\Leftrightarrow$ research activities
 - IBs: spinning of IPO's (on hot IPO shares)
 - Accounting firms: auditing and consulting
 - Credit Rating agencies: credit ratings and ancillary consulting services
- ✓ Financial Regulation may reduce them
 - Sabanes-Oxley Act (2002): addressed some of the concerns w.r.t. the auditing process (ex: accounting Vs audit)
 - Global Legal Settlement (2002): prohibited spinning and has drawn a line between research and underwriting activities of investment banks